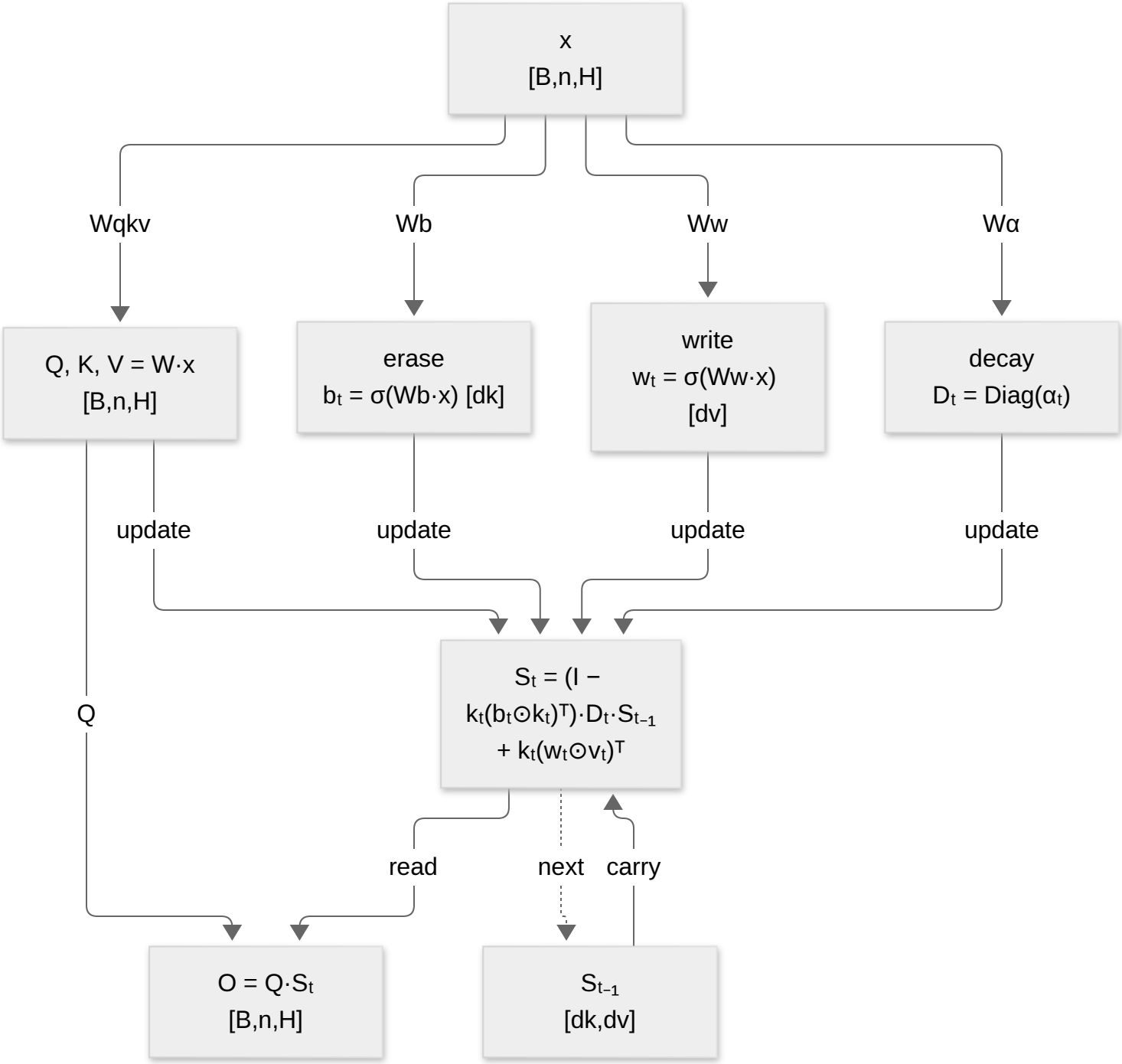

x
[B,n,H]
Wqkv
Wb
Ww
Wα
Q, K, V = W·x
[B,n,H]
erase
$b_t = \sigma(Wb \cdot x)$ [dk]
write
$w_t = \sigma(Ww \cdot x)$
[dv]
decay
$D_t = Diag(\alpha_t)$
update
update
update
update
Q
$S_t = (I - k_t(b_t \odot k_t)^T) \cdot D_t \cdot S_{t-1} + k_t(w_t \odot v_t)^T$
read
next
carry
$O = Q \cdot S_t$
[B,n,H]
S_{t-1}
[dk,dv]